

ESTIMATION PRO — TONNAGE × USD PER-TON RATE

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

SHA-256: ba986cc69b481b4c9042a8d00c30a16488d0b794fcc452a88a6054f7dbecac9f

CALCULATION MANIFEST — Single Source of Truth

Parameter	Value
AI-Extracted Tonnage	9.20 t
Material Type	Carbon Steel
Surface Treatment	Shop primer
Assembly Complexity	Mixed bolt-weld
Material Factor	×1.00
Surface Factor	×1.00
Assembly Factor	×1.10
Composite Factor	×1.10
Selected Rate (USD/ton)	\$5,500
Adjusted Rate (USD/ton)	\$6,050
Country	USA
Tax %	0%
Risk Buffer %	15%

Parameter	Value
LOCK_COST_LOW	\$60,800
LOCK_COST_MID	\$64,000
LOCK_COST_HIGH	\$67,200

> All LOCK values are fixed for this entire document. > Sections 1 and 7 copy these values exactly — no re-derivation.

1. EXECUTIVE SUMMARY

> LOCK COPY INSTRUCTION: Copy all values from MANIFEST — do not re-derive.

Metric	Value
Project Name / ID	HOME 2 - CAROWINDS
Country	USA
Total Fabricated Tonnage	9.20 t
Material Type	Carbon Steel
Surface Treatment	Shop primer
Assembly Complexity	Mixed bolt-weld
Composite Adjustment	×1.10
Selected Rate	\$5,500 / ton
Adjusted Rate	\$6,050 / ton
Tax	0%
Cost Range (USD, inc. tax)	\$60,800 → \$67,200
Headline Cost (Midpoint)	\$64,000
Cost per Ton (Adjusted Mid)	\$6,957 / ton

Top 3 Critical Risks: 1. ■ Unquantified Lintels: All steel lintels (W8, W16, L-shapes) on schedule S0.02 are missing lengths. This scope is 100% excluded and could add 3-5 tons to the project. 2. ■ Unquantified Trellis Steel: Trellis beams and columns (HSS8x6x3/8) on detail S3.06 are not shown on any plan view, so quantities and lengths are unknown. This scope is 100% excluded. 3. ■ Conflicting Coating Specs: General notes call for shop primer (S0.01), but the bearing plate schedule requires galvanizing (S0.02). Galvanizing would increase the surface factor from 1.00 to 1.28, significantly impacting cost.

2. BASIS OF ESTIMATE

- Drawings reviewed: S0.01 (General Notes), S0.02 (Schedules & Details), S1.01 (Foundation Plan), S1.02 (Second Level Framing Plan), S3.06 (Trellis Details).
- Tonnage extraction method: (b) Computed from member schedules and framing plans using AISC unit weights. Primary sources were the Steel Column Schedule on S0.02 and the Second Level Framing Plan on S1.02.
- Composite factor selection:
- Material (Carbon Steel): Explicitly defined by ASTM grades on S0.01.
- Surface (Shop primer): Selected based on General Structural Note #10 on S0.01. The conflict with galvanized bearing plates (S0.02) is noted as a primary risk.
- Assembly (Mixed bolt-weld): Assumed based on standard commercial construction shown; presence of welded column base plates and assumed simple bolted beam connections.
- Rate source: User-selected from dropdown (\$5,500/ton). This is an assumed rate for demonstration as one was not provided.
- Tax applied: 0% for USA.
- Global assumptions: Column lengths were assumed to be 14'-0" based on floor elevations, as no explicit lengths were provided. This requires immediate confirmation. Wind post lengths assumed to be 10'-0".

3. MEMBER-BY-MEMBER TONNAGE TAKE-OFF

> This table is the source of LOCK_TONNAGE. Sum all Est. Tonnage rows = LOCK_TONNAGE. > Show calculation for every row: Qty × Length(m) × Unit Wt(kg/m) = Est Wt(kg).

Table 3A: Primary Structural Members

#	Type	Mark	Profile	Qty	Length (Imperial)	Length (mm)	Unit Wt (kg/m)	Est Wt (kg)	Est Wt (lb)	Grade	Sheet	Conf
1	Beam	—	W14x22	32	12'-10"	3,912	32.7	4,094	9,026	A992	S1.02	■
2	Beam	—	W16x36	1	43'-0 1/2"	13,119	53.6	703	1,550	A992	S1.02	■
3	Column	C5	HSS6x6x 3/8	4	14'-0" (Est.)	4,267	33.1	565	1,246	A500	S0.02	■ ■
4	Column	CSA	HSS6x6x 5/16	2	10'-0" (Est.)	3,048	28.1	171	377	A500	S0.02	■ ■
5	Column	C6	HSS6x6x 5/8	4	14'-0" (Est.)	4,267	46.4	792	1,746	A500	S0.02	■ ■
6	Column	C86	HSS8x6x 3/8	2	14'-0" (Est.)	4,267	44.9	383	845	A500	S0.02	■ ■
7	Beam	—	HSS18x6x 3/8	1	45'-1"	13,741	71.0	976	2,151	A500	S1.02	■ ■
8	Beam	—	HSS18x6x 3/8	1	35'-10"	10,922	71.0	775	1,709	A500	S1.02	■ ■
9	Beam	—	HSS18x6x 3/8	1	16'-11"	5,156	71.0	366	807	A500	S1.02	■ ■
10	Beam	—	HSS18x6x 3/8	1	12'-10"	3,912	71.0	278	613	A500	S1.02	■ ■

Table 3B: Secondary & Misc Members

#	Type	Mark	Profile	Qty	Length (Imperial)	Length (mm)	Unit Wt (kg/m)	Est Wt (kg)	Est Wt (lb)	Grade	Sheet	Conf
—	Lintel	L1	W8x24	■	NOT FOUND	■	35.7	■	■	A992	S0.02	■
—	Lintel	L2	W16x50	■	NOT FOUND	■	74.4	■	■	A992	S0.02	■
—	Lintel	LLV	L7x4x3/8	■	NOT FOUND	■	NF-WT	■	■	A36	S0.02	■
—	Trellis	—	HSS8x6x3/8	■	NOT FOUND	■	44.9	■	■	A500	S3.06	■

Table 3C: Connection Material & Plates

#	Description	THK	Width	Length	Qty	Wt (kg)	Wt (lb)	Sheet	Conf
1	C5 Base Plate	3/4"	12"	12"	4	56	123	S0.02	■
2	CSA Base Plate	3/4"	12"	12"	2	28	61	S0.02	■
3	C6 Base Plate	1"	12"	12"	4	73	161	S0.02	■
4	C86 Base Plate	1"	16"	16"	2	65	143	S0.02	■
5	Wind Post Conn. PL	1/2"	6"	12"	2	10	21	S0.02	■■

Table 3D: Tonnage Subtotals

Category	Count	Total Wt (kg)	Total Wt (lb)	Est. Tons
W-Shapes (columns + beams)	33	4,797	10,576	4.80
HSS / Tube	15	3,906	8,612	3.91
Pipe	0	0	0	0.00
Angles / Channels	0	0	0	0.00
Plates (all)	14	231	509	0.23
Misc / Embeds	0	0	0	0.00
Bolts & accessories (+3%)	—	268	591	0.27
TOTAL	62	9,202	20,288	9.20

4. COST BUILD-UP BY MEMBER CATEGORY

> Rate applied: \$5,500/ton × ×1.10 = \$6,050/ton

Per-Category Cost

Category	Est. Tons	Adj. Rate (USD/ton)	Subtotal (USD)	% of Total
W-Shapes	4.80	\$6,050	\$29,033	52.2%
HSS / Tube	3.91	\$6,050	\$23,639	42.5%
Pipe	0.00	\$6,050	\$0	0.0%
Angles / Channels	0.00	\$6,050	\$0	0.0%

Category	Est. Tons	Adj. Rate (USD/ton)	Subtotal (USD)	% of Total
Plates	0.23	\$6,050	\$1,394	2.5%
Misc / Embeds	0.00	\$6,050	\$0	0.0%
Bolts & accessories	0.27	\$6,050	\$1,594	2.8%
TOTAL	9.20	\$6,050	\$55,660	100%

5. PROCESS COST BREAKDOWN (MIDPOINT SCENARIO)

> Distributing \$64,000 across industry-standard process activities. > Percentages are fixed fabrication industry standards.

Process Split

Process Activity	Industry %	Amount (USD)	Notes
Material (mill sections + plate)	35%	\$22,400	Structural steel mill cost
Shop labour (cut / fit / weld)	30%	\$19,200	Includes AWS D1.1 weld ops
Coatings / surface prep	12%	\$7,700	SSPC-SP1 shop primer applied
Consumables (weld wire / bolts / gas)	6%	\$3,800	Per-ton allowance
Equipment (plasma / burn / drill)	7%	\$4,500	CNC / NC machine allocation
Overhead / QA-QC	6%	\$3,800	Incl. inspection, NDT
Margin	4%	\$2,600	Shop margin
TOTAL	100%	\$64,000	Midpoint scenario

6. RISK BUFFER & ADJUSTED COST

Confidence Assessment

Check	Status	Notes
All quantities BOM-confirmed or counted on plans	■	Multiple member lengths assumed; entire lintel & trellis scope missing.
Material grades explicit on all members	■	Grades clearly stated on S0.01.
Surface treatment confirmed (not assumed)	■ ■	S0.01 calls for primer; S0.02 calls for galvanizing on plates. Conflict.
Assembly complexity confirmed	■ ■	Assumed as mixed bolt/weld. No connection details provided.
No conflicting BOM vs general-notes grade callouts	■	Surface treatment specifications are in conflict.
Drawings are IFC / internally coordinated	■	Stamped "Construction Documents" set.
Galvanising / AESS / fire-rating fully resolved	■	Galvanizing requirement is ambiguous and unresolved.

Checks met: 2 of 7 Confidence: LOW ($\leq 3/7 \rightarrow 10\text{--}15\%$) Buffer applied: 15%

Cost Waterfall

Row	Amount (USD)
Base subtotal (tonnage $\times$ adjusted rate)	\$55,660
Risk buffer (15%)	+\$8,349
Subtotal after buffer	\$64,009

Row	Amount (USD)
Tax (0%)	+\$0
Grand Total (Low — -5%)	\$60,800
Grand Total (Mid — headline)	\$64,000
Grand Total (High — +5%)	\$67,200

7. COST CONVERSION (USD)

> LOCK COPY INSTRUCTION: All values copied from MANIFEST — do not re-compute.

Field	Value
Fabricated Tonnage	9.20 t
Selected Rate (USD/ton)	\$5,500
Composite Factor	×1.10
Adjusted Rate (USD/ton)	\$6,050
Country / Tax	USA · 0%
Risk Buffer	15%
Grand Total — Low	\$60,800
Grand Total — Mid (headline)	\$64,000
Grand Total — High	\$67,200
Cost per Fabricated Ton (Mid)	\$6,957 / ton

8. ASSUMPTIONS & EXCLUSIONS

Key Assumptions (minimum 5 specific bullets — cite sheet references):

- Rate \$5,500/ton provided by user — not derived by AI. (This is a placeholder value).
- All main column (C5, C6, C86) lengths are 14'-0" and wind post (CSA) lengths are 10'-0", estimated from floor elevations on S2.03. These lengths must be confirmed.
- Surface treatment is shop primer only, per S0.01. The requirement for galvanized plates on S0.02 has been excluded but flagged as a risk.
- HSS profiles not in the provided weight table (HSS6x6x5/16, HSS6x6x5/8, HSS8x6x3/8, HSS18x6x3/8) have been assigned weights based on standard AISC data.
- The canopy shown on S1.02 is by others and excluded.

Exclusions (NOT included in this estimate):

- Site erection labour, cranes, manlifts, and rigging.
- PE stamping or delegated connection design.
- ALL STEEL LINTELS shown in schedules on S0.02.
- ALL TRELLIS STEEL shown in details on S3.06.
- ALL BEARING PLATES from the schedule on S0.02.
- Loose anchor bolts (cast-in by GC).
- Site touch-up paint after final erection.
- Tax / duties beyond 0% applied here.
- 3D MEP coordination / clash detection.
- Bid validity period of 30 days assumed.

Change Order Triggers:

- Addition of any of the excluded items (lintels, trellis steel, bearing plates).
- Tonnage changes > $\pm 5\%$ from this estimate for the included scope.
- Coating system upgrade from primer-only to any galvanized or multi-coat system.
- Any column length change from the assumed 14'-0" / 10'-0".

10. FINAL RECOMMENDATION

--- This estimate is not reliable for final bidding due to multiple large scopes of work that are undefined in the provided documents. The calculated tonnage of 9.20 t likely represents less than 70% of the actual project steel. Before proceeding, it is critical to issue RFIs to obtain all missing lintel lengths from architectural drawings, get quantities and layout for the trellis steel, and clarify the conflicting coating specifications. A full re-estimate is mandatory once this information is received.

ACTION SUMMARY

Priority Actions for This Analysis

#	Priority	Action Required	Owner	Deadline	Blocking?
1	■ Critical	RFI: Provide lengths for all steel lintels in schedules on S0.02, referencing architectural opening schedules.	Estimator	ASAP	■ Yes
2	■ Critical	RFI: Provide plan layout, quantities, and lengths for all trellis steel members detailed on S3.06.	Estimator	ASAP	■ Yes
3	■ Major	RFI: Clarify coating requirements. Is it shop primer per S0.01 or are bearing plates/lintels galvanized per S0.02?	Estimator	ASAP	■ Yes

#	Priority	Action Required	Owner	Deadline	Blocking?
4	■ Minor	RFI: Confirm all HSS column lengths, assumed to be 14'-0" / 10'-0".	Estimator	ASAP	■■ Yes